



## A Study of Li-Ion Cells Operated to 4.5 V and at 55°C

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The long-term cycling behavior of 24 promising electrolyte blends were systematically studied in LaPO<sub>4</sub>-coated Li(Ni<sub>0.4</sub>Mn<sub>0.4</sub>Co<sub>0.2</sub>)O<sub>2</sub>/graphite pouch type Li-ion cells tested to 4.5 V at 55°C. Capacity fade during cycling, charge-transfer resistance ( $R_{ct}$ ) before and after cycling as well as gas evolution during formation and also during cycling were examined and compared head-to-head. Of all the electrolytes tested, triallyl phosphate containing electrolytes including 2% vinylene carbonate + 2% triallyl phosphate in 1 M LiPF<sub>6</sub> sulfolane:ethyl methyl carbonate and 2% prop-1-ene sultone + 2% triallyl phosphate in 1M LiPF<sub>6</sub> ethylene carbonate:ethyl methyl carbonate electrolytes showed the best capacity retention, the least impedance growth and manageable amounts of gas evolution during long-term cycling. Pyridine boron trifluoride -based additives also showed excellent cycling performance but cells with those additives had higher gas evolution during cycling. Cells containing fluorinated electrolytes had similar cycling performance to cells containing 2% prop-1-ene sultone in ethylene carbonate:ethyl methyl carbonate electrolyte but with much more gas evolution and higher impedance after long-term cycling. The results suggest that electrolytes containing the additives triallyl phosphate or pyridine boron trifluoride may be promising for high voltage Li-ion cells at elevated temperature.

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There is an ever increasing demand for higher energy density Li-ion batteries.<sup>1</sup> Simply charging conventional cathode materials (e.g. LiCO<sub>2</sub>) to higher voltage or employing high voltage cathode materials can achieve higher energy density, but this is normally at the expense of lifetime.<sup>2,3</sup> The poor cycling stability is mainly caused by the electrochemical oxidation reactions between the charged positive electrode and electrolyte at high voltages.<sup>4,5</sup> The parasitic reaction between the charged electrode and the electrolyte at high potentials results in gas evolution,<sup>6,7</sup> severe impedance growth,<sup>8,9</sup> low coulombic efficiency (CE) and poor capacity retention.<sup>10</sup> The formation of a rock-salt layer on the surface of the positive electrode materials may also contribute to the impedance growth and capacity loss of the cells.<sup>11</sup>

There are considerable numbers of publications that focus on methods including surface coating,<sup>12–16</sup> electrolyte additives<sup>17–22</sup>, as well as new electrolyte solvents<sup>23–28</sup> which either form a physical protective layer between the electrolyte and the positive electrode or broaden the voltage window of the electrolyte to stabilize the positive electrode/electrolyte interphase. Most academic researchers focused on one of the above methods and performed tests in coin-type of half cells using a Li metal counter electrode. For example, Tarnopolskiy et al.<sup>29</sup> studied the impact of over 40 electrolyte additives in LiNi<sub>0.5</sub>Mn<sub>1.5</sub>O<sub>4</sub>/Li half cells cycled between 3.5 and 4.95 V at C/2 and found only succinic anhydride decreased the self-discharge and increased the capacity retention during cycling. Lee et al.<sup>30</sup> investigated the AlF<sub>3</sub> layer coating in Li[Ni<sub>0.8</sub>Co<sub>0.15</sub>Al<sub>0.05</sub>]O<sub>2</sub>/Li half cells and found that the AlF<sub>3</sub> coating greatly suppressed the impedance growth and improved the cycling stability. Zhang and Amine et al.<sup>31</sup> showed that LiNi<sub>0.5</sub>Mn<sub>1.5</sub>O<sub>4</sub> (LNMO)/Li coin-type cells with all-fluorinated electrolytes had higher lithium ion conduction and superior anodic stability at elevated temperatures compared to conventional EC:EMC-based electrolytes. However, when cycled in LNMO/graphite chemistry, cells containing these all-fluorinated electrolytes lost more than 50% of their initial capacity after only 100 cycles at C/3 and 3.5–4.9 V at 55°C.<sup>3,2</sup> This is because in a full cell, the amount of active lithium is limited by the positive electrode. Any parasitic reactions (eg. electrolyte oxidation, transition metal dissolution, salt consumption and solid electrolyte interphase (SEI) growth on both positive and negative electrodes) which causes the loss of active lithium will decrease

coulombic efficiency and shorten cycle-life time.<sup>33</sup> By contrast, in a half cell, this lithium loss will be recovered from the excess lithium foil and will not appear as capacity loss. Therefore, it is not surprising that the cycling performance in LiNi<sub>0.5</sub>Mn<sub>1.5</sub>O<sub>4</sub>/graphite cells with all-fluorinated electrolytes was dramatically improved with a lithium reservoir.<sup>3,4</sup>

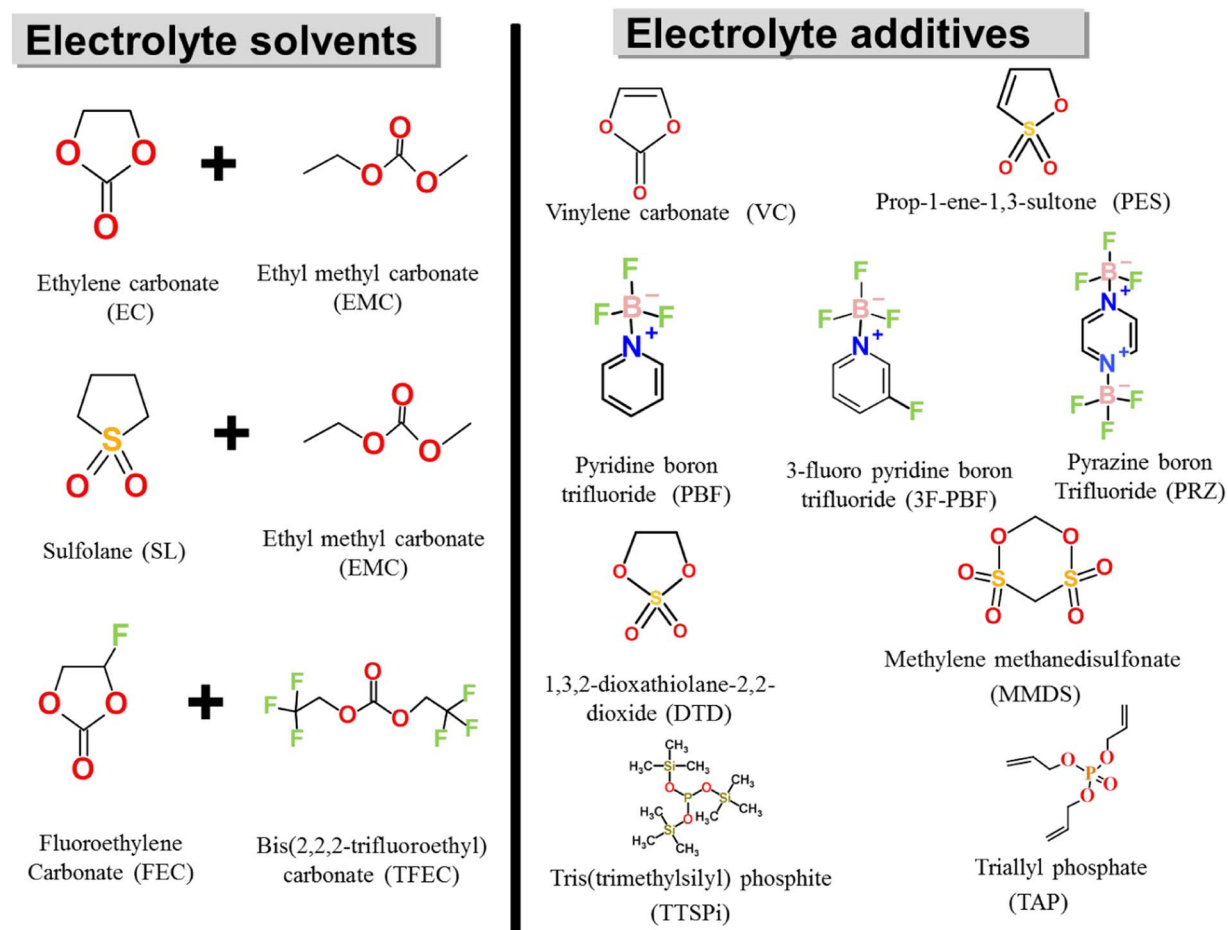
Recently, the impact of electrolyte additives, coatings as well as novel solvents in NMC111/graphite and NMC422/graphite pouch type Li-ion cells has been studied. Wang et al.<sup>35</sup> studied over 110 electrolyte additive blends in NMC111/graphite cells tested to 4.2 V and found that an electrolyte consisting of 2% prop-1-ene sultone (PES), 1% ethylene sulfate (DTD) and 1% tris-trimethylsilyl phosphate (TTSPi), also called “PES 211”, showed the highest CE, lowest charge end-capacity slippage, lowest gas evolution and excellent impedance control during cycling on an Ultra High Precision Charger (UHPC). Nelson et al.<sup>9</sup> and Ma et al.<sup>36,37</sup> showed that PES 211 could greatly suppress the impedance growth and parasitic heat flow at high potentials in NMC422/graphite pouch cells. Nie et al.<sup>38–40</sup> showed that pyridine boron trifluoride (PBF)-based electrolyte additives had clear benefits in NMC442/graphite cells such as limiting the electrolyte oxidation as well as the impedance growth during storage and long-term cycling. Xia et al.<sup>41</sup> showed that triallyl phosphate could polymerize and form thick SEI films at the surfaces of both graphite and NMC442 electrodes which led to better cycling stability at high voltages.

The impact of coatings on NMC 422/graphite pouch cells has also been studied. Nelson et al.<sup>42</sup> and Xia et al.<sup>43</sup> found that a LaPO<sub>4</sub> coating on NMC442 had little if any benefit below 4.5 V. At 4.5 V or above, the LaPO<sub>4</sub>-coated cells had a significantly smaller voltage drop during elevated temperature storage than the uncoated cells except when fluorinated electrolytes were used.<sup>44</sup> Some alternative electrolyte formulations with high anodic stability were studied in NMC442/graphite pouch cells as well. For example, Xia et al.<sup>45,46</sup> showed that sulfolane (SL)-based electrolytes could provide similar or better cycling and storage performance compared to state-of-the-art electrolytes with additives in NMC442/graphite pouch cells tested to 4.5 V. Xia et al.<sup>47</sup> also showed that fluorinated solvent mixtures containing fluoroethylene carbonate (FEC) and bis(2,2,2-trifluoroethyl) carbonate (TFEC) with PES as an electrolyte additive outperformed all non-fluorinated electrolytes with all additives during continuous charge-discharge cycling tests to 4.5 V and at 40°C in NMC442/graphite pouch cells. Although these electrolyte additives and/or solvents illustrated superior capability during short term cycling, a head-to-head comparison of the cycle stability during long-term cycling was not addressed.

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**Figure 1.** Electrolyte formulations as well as structural information for the electrolyte solvents and additives studied in this paper.

In this paper, 24 electrolyte blends with different electrolyte solvents and/or electrolyte additives were carefully studied and compared head-to-head in  $\text{LaPO}_4$ -coated NMC442/graphite pouch cells tested to 4.5 V and 55°C. Figure 1 shows the chemical structures of these selected electrolyte solvents and electrolyte additives. The reasons for choosing these additives and solvents are based on Refs. 35–47, which show these electrolyte blends have superior performance during cycling or storage. Long-term cycling stability, impedance before and after cycling as well as cell swelling before and after long-term cycling were measured and were compared with 2% PES in EC:EMC-based electrolyte.

### Experimental

**Chemicals.**—1 M  $\text{LiPF}_6$  in ethylene carbonate (EC): ethyl methyl carbonate (EMC) (3:7 wt% ratio, BASF, 99.99%) was used as the control electrolyte. The other two electrolyte systems, 1 M  $\text{LiPF}_6$  SL:EMC (3:7 wt% ratio) and 1 M  $\text{LiPF}_6$  FEC:TFEC (1:1 wt% ratio), were chosen based on our previous publications.<sup>45,47</sup> Additives were included in the three electrolyte systems in a range of 0.5 to 5 wt%. These additives included vinylene carbonate (VC), PES, methane methylenedisulfonate (MMDS), propanediol cyclic sulfate (trimethylene sulfate - TMS), DTD, TTSPi, lithium bis(oxalate)borate (LiBOB), triallyl phosphate (TAP), PBF, pyrazine boron trifluoride (PRZ) as well as 3-fluoro pyridine boron trifluoride (3F-PBF). The purities and the suppliers of the solvents and additives used are listed in Table I.

**Pouch cells, formation and degassing.**—The 402035-size pouch cells employed in this study were  $\text{LaPO}_4$ -coated NMC442/graphite cells with a capacity of 180 mAh balanced for 4.7 V operation. The

positive NMC442 active material was coated with 3 wt% of  $\text{LaPO}_4$ , which appears as nanoparticles on the NMC442 particle surfaces. The NMC442 was prepared by Umicore Co. and the  $\text{LaPO}_4$  coating was applied by 3M Co. The pouch cells were manufactured by Li-Fun Technology (Zhuzhou, China), sealed without electrolyte in China and then shipped to our laboratory in Canada. Before filling with electrolyte, the cells were cut just below the heat seal and dried at 80°C under vacuum for 14 h to remove most of the residual water. The cells were then transferred immediately to an argon-filled glove box for electrolyte filling and vacuum sealing. The NMC442/graphite pouch cells were filled with 0.75 mL of electrolyte (0.90 g for EC:EMC, 0.87 g for SL:EMC and 1.15 g for FEC:TFEC electrolyte because the density of EC:EMC electrolyte is 1.20 g/mL, compared to 1.17 g/mL for SL:EMC and 1.53 g/mL for FEC:TFEC). After filling, cells were vacuum-sealed with a compact vacuum sealer (MSK-115A, MTI Corp.). First, cells were placed in a temperature box at 40.0°C where they were held at 1.5 V for 24 hours, to allow for the completion of wetting. Then, cells were charged at C/20 to 3.5 V and were held at 3.5 V for 1 hour. This step is called formation step 1. After formation step 1, cells were transferred into the glove box, cut open just below the seal to release any gas generated during the charge to 3.5 V and vacuum sealed again. These cells were then charged from 3.5 V at C/20 to 4.5 V. This step is called formation step 2. After formation step 2, the cells were transferred into the glove box, cut open to release gas generated and then vacuum sealed again. These degassing voltages were selected based on the in-situ gas evolution experiments that show that most of the gas evolves in the formation step at voltages below 3.5 V and above 4.3 V.<sup>6</sup> After the two degassing processes, cells were then discharged to 3.8 V where the cell volume and impedance spectra were measured.

**Table I.** List of abbreviations, solvents or additives used and purity, as well as their suppliers.

| Solvents or Additives                                                                       | Purity by weight | Suppliers                               |
|---------------------------------------------------------------------------------------------|------------------|-----------------------------------------|
| FEC - fluoroethylene carbonate                                                              | 99.94%           | BASF                                    |
| TFEC - bis(2,2,2-trifluoroethyl) carbonate                                                  | 99.80%           | HSC Corporation (Jiangsu, China)        |
| SL - sulfolane                                                                              | 99.99%           | Novolyte Technologies                   |
| EMC - ethyl methyl carbonate                                                                | 99.99%           | BASF                                    |
| VC - vinylene carbonate                                                                     | 99.8%            | BASF                                    |
| PES - prop-1-ene,1,3-sultone                                                                | 98.7%            | Lianchuang Medicinal Chemistry Co., Ltd |
| DTD - 1,3,2-dioxathiolane-2,2-dioxide— also called ethylene sulfate                         | 98%              | Sigma-Aldrich                           |
| TMS - 1,3,2-dioxathiane 2,2-dioxide— also called trimethylene sulfate                       | 98%              | Sigma-Aldrich                           |
| MMDS - 1,5,2,4-dioxadithiane-2,2,4,4-tetraoxide – also called methylene methane disulfonate | > 98.7%          | Tinci Materials Technology              |
| LiBOB - Lithium bis-oxalatoborate                                                           | ≥ 99.9%          | Rockwood Lithium                        |
| TTSPi - tris(trimethylsilyl) phosphite                                                      | ≥ 95.0% (GC)     | Sigma-Aldrich                           |
| TAP - triallyl phosphate                                                                    | > 94.0% (GC)     | TCI America                             |
| ES -1,3,2-Dioxathiolan-2-oxide – also called ethylene sulfite                               | ≥ 99.0%          | Sigma-Aldrich                           |
| pyridine boron trifluoride (PBF)                                                            | > 99% (GC)       | Synthesized in our lab                  |
| pyrazine boron trifluoride (PRZ)                                                            | > 99% (GC)       | Synthesized in our lab                  |
| 3-fluoro pyridine boron trifluoride (3F-PBF)                                                | > 99% (GC)       | Synthesized in our lab                  |

**Ex-situ gas measurement.**—Ex-situ gas measurements were made by suspending pouch cells from a fine wire “hook” attached under a Shimadzu balance (AUW200D).<sup>48</sup> The pouch cells were immersed in a beaker of de-ionized “nanopure” water (18 MΩ) that was at 20. ± 1°C for measurement. Before weighing, all cells were charged or discharged to 3.80 V. The changes in the weight of the cell suspended in fluid, before, during and after testing are directly related to the volume changes by the change in the buoyant force. The change in weight of a cell,  $\Delta w$ , suspended in a fluid of density,  $\rho$ , is related to the change in cell volume,  $\Delta v$ , in mL by

$$\Delta v = -\Delta w/(\rho g) \quad [1]$$

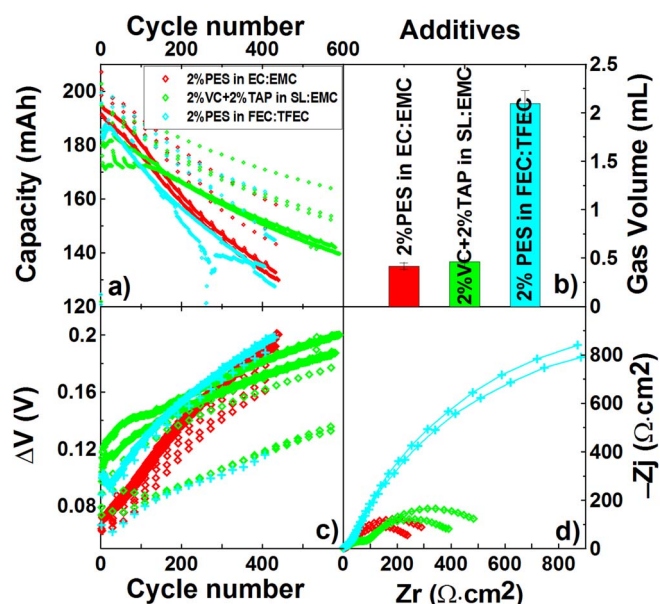
where the weight was measured in milliNewtons, the density in g/mL and  $g$  is the acceleration due to gravity in m/sec<sup>2</sup>. It is important to realize that  $w/g$  is what a balance reports as the mass in grams.

**Electrochemical impedance spectroscopy.**—Electrochemical impedance spectroscopy (EIS) measurements were conducted on NMC442/graphite pouch cells after formation and after cycling.<sup>49</sup> Cells were charged or discharged to 3.80 V before they were moved to a 10. ± 0.1°C temperature box. The potential of 3.8 V was chosen for the EIS measurements because at this voltage both graphite and NMC electrodes are at about 50% state of charge and are stable. The temperature 10°C was chosen because the impedance at 10°C is much larger than in the same cells tested at 30°C.<sup>50</sup> Therefore it is easier to differentiate between the impact of additives when the impedance spectra are measured at 10°C. Alternating current (AC) impedance spectra were collected with ten points per decade from 100 kHz to 10 mHz with a signal amplitude of 10.0 mV at 10. ± 0.1°C. A Biologic VMP-3 was used to collect these spectra. The experimental setup did not allow for reproducible solution resistance measurements due to cable and connector impedance. Therefore, all impedance spectra were manually shifted to zero on the real axis at the highest frequency measured.

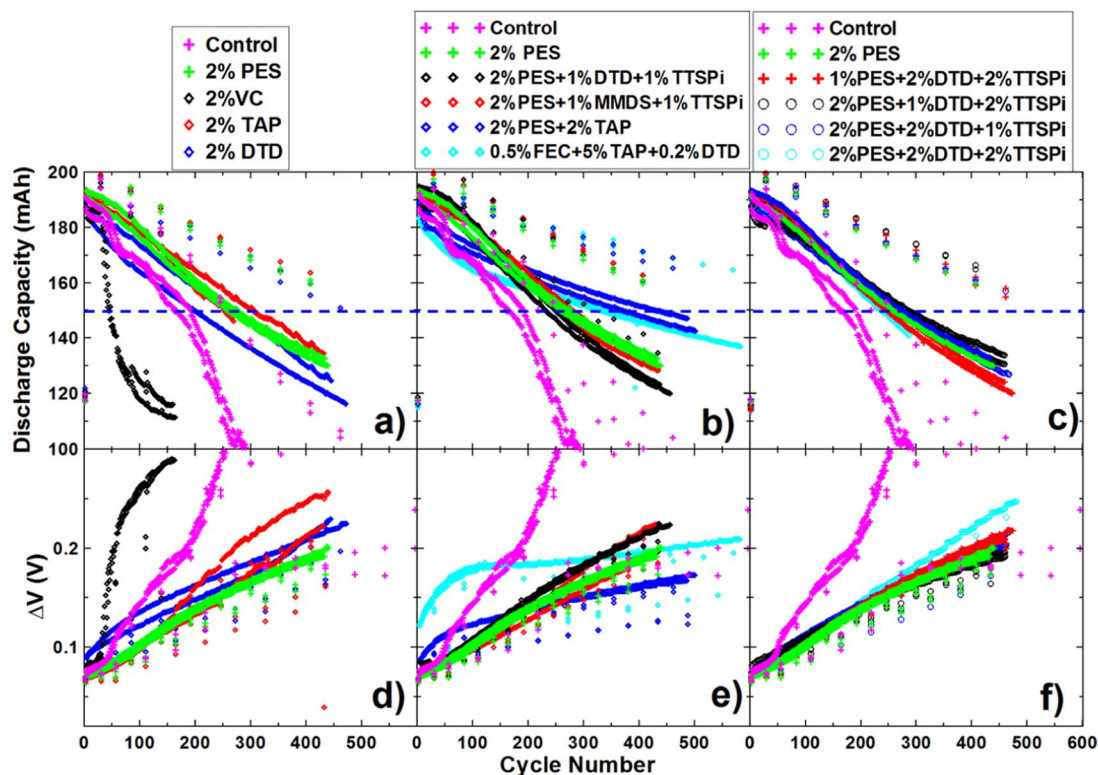
**Long-term cycling.**—After formation, cells were continuously cycled between 2.8 V and 4.5 V at 55 ± 0.5°C using currents corresponding to C/2.3 (80 mA). The cells were clamped between two rectangular silicone rubber pieces (McMaster-Carr part number 5508T45) placed in the cell holder with aluminum shims added to ensure firm pressure.<sup>45</sup> With the clamps in place, gas is pushed to the edges of the pouch where there is ample space for it and the clamps maintain stack pressure on the cell. A low rate C/20 (9 mA) cycle was included every 25 cycles to gauge what fraction of the capacity loss was due to impedance growth during the high rate cycling.

## Results and Discussion

Figure 2 shows typical data collected during long-term cycling experiments. Three electrolyte systems including 2% PES in EC:EMC 3:7, 2% VC + 2% TAP in SL:EMC 3:7 and 2% PES in FEC:TFEC 1:1 were chosen for comparison. Figure 2a shows discharge capacity vs. cycle number of NMC442/graphite cells containing these three electrolyte systems. Figure 2a shows cells containing 2% PES in FEC:TFEC electrolyte had similar capacity retention to cells containing 2% PES in EC:EMC electrolyte. Figure 2a shows cells containing 2% VC + 2% TAP in SL:EMC electrolyte had lower capacity at the beginning due to high impedance but had better capacity retention than cells containing 2% PES in EC:EMC electrolyte. Figure 2b shows the volume change during long-term cycling at 55°C for the same cells. Since these cells were cycled with clamps in place to maintain pressure on the electrode stack, the cycling performance was not affected by small volumes of gas evolution. Figure 2b shows that



**Figure 2.** a) Discharge capacity vs. cycle number during long-term cycling at 55°C (2.8 to 4.5 V) with a current density of C/2.3 (80 mA), b) gas evolution during long-term cycling at 55°C (c)  $\Delta V$  vs. cycle number during long-term cycling and d) Nyquist plots, measured at 3.8 V and 10°C, after long-term cycling for LaPO<sub>4</sub>-coated NMC442/graphite pouch cells with different electrolytes as indicated. Figure 2 gives example data showing the methods used in this paper.



**Figure 3.** a, b, c) Discharge capacity and d, e, f)  $\Delta V$ , all plotted vs cycle number for  $\text{LaPO}_4$ -coated NMC442/graphite pouch cells containing different electrolytes as indicated. All cells in Figure 3 are in 1M  $\text{LiPF}_6$  EC:EMC 3:7. The cycling was between 2.8 and 4.5 V at C/2.3 (80 mA) and 55°C using a continuous cycling protocol.

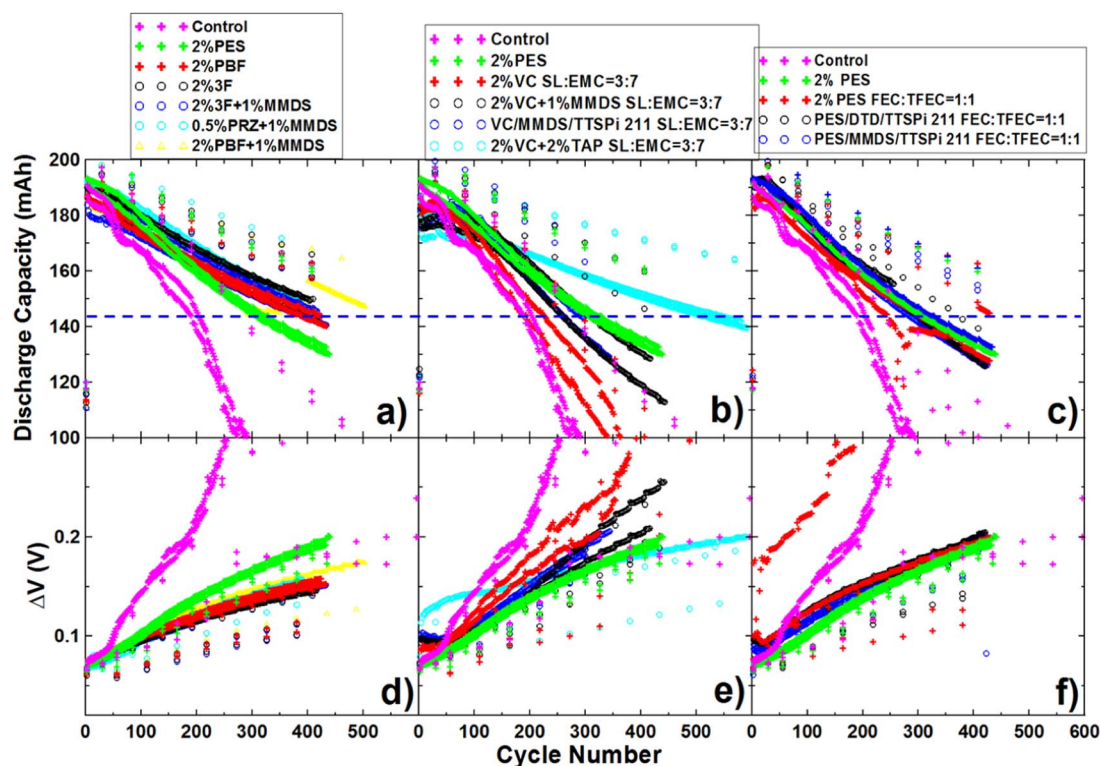
cells containing 2% VC + 2% TAP in SL:EMC electrolyte produced about 0.4 mL gas, similarly to cells containing 2% PES in EC:EMC electrolyte. Cells containing 2% PES in FEC:TFEC electrolyte produced about 2.1 mL gas which is simply too much for a pouch cell with a total volume of  $\sim 2.2$  mL. Figure 2c shows the difference between the average charge and discharge potential ( $\Delta V$ ) vs. cycle number for the same cells shown in Figure 2a. A smaller value of  $\Delta V$  means there was smaller polarization and thus smaller overall cell impedance (DC impedance).<sup>51</sup> Figure 2c shows cells containing 2% PES in EC:EMC or 2% PES in FEC:TFEC electrolyte had more rapid impedance growth during cycling than cells containing 2% VC + 2% TAP in SL:EMC electrolyte. Figure 2d shows the impedance spectrum measured after long-term cycling experiments. The EIS measurements were made at 10°C with a cell voltage of 3.80 V. The diameter of the semicircle represents the sum of the charge-transfer resistances,  $R_{ct}$ , at both the positive and negative electrodes. Smaller values of  $R_{ct}$  are desired for these cells cycled during the same period of time. Figure 2d shows that cells containing 2% VC + 2% TAP in SL:EMC and 2% PES in FEC:TFEC electrolytes had higher impedance than cells containing 2% PES in EC:EMC electrolyte after cycling.

Figures 3a, 3b and 3c show the discharge capacity vs. cycle number for the NMC442/graphite pouch cells containing different additives and their combinations as indicated. Figures 3d, 3e and 3f show the  $\Delta V$  vs. cycle number for the same cells shown in Figures 3a, 3b and 3c, respectively. All cells in Figure 3 had 1M  $\text{LiPF}_6$  EC:EMC 3:7 base electrolyte. Figure 3 clearly shows the impact of electrolyte additives on the cycling performance in EC-based electrolyte. Figures 3a and 3d show that cells containing control lost 20% of their initial capacity after 200 cycles while cells containing 2% VC reached 20% capacity loss after only 50 cycles due to the unstable SEI formed by VC at high voltage and/or high temperatures.<sup>36,52</sup> Figures 3a and 3d show that cells containing 2% PES, 2% DTD or 2% TAP had similar capacity retention during cycling and better cycling performance than control cells. Figures 3b and 3c show that cells containing any combinations

of PES/DTD (or MMDS)/TTSPi in EC:EMC electrolyte had similar cycling performance to cells containing only 2% PES in EC:EMC electrolyte which was very different from the results reported by Ma et al.<sup>36</sup> and Nelson et al.<sup>9</sup> which showed that these combinations have much better cycling performance than 2% PES at 40°C in NMC442 cells **without the  $\text{LaPO}_4$  coating**. Figure 3b shows that cells containing 2% PES + 2% TAP or 0.5% FEC + 5% TAP + 0.2% DTD had much better capacity retention and less impedance growth than cells containing 2% PES in EC:EMC electrolyte or combinations of PES/DTD(MMDS)/TTSPi.

Figure 4 shows the capacity versus cycle number for cells with PBF-based electrolyte additives (Figures 4a and 4d),<sup>38–40</sup> sulfolane-based electrolytes (Figures 4b and 4e)<sup>45</sup> and fluorinated electrolytes (Figures 4c and 4f)<sup>4,8</sup> all compared with 2% PES in EC:EMC 3:7. Figures 4a and 4d show that these PBF-based electrolyte additive blends had better capacity retention and less impedance growth than 2% PES in EC:EMC electrolyte during cycling. Figures 4b and 4e show that cells containing 2% VC, 2% VC + 1%MMDS and 2% VC + 1% MMDS + 1%TTSPi in SL:EMC electrolyte had worse capacity retention than cells containing 2% PES in EC:EMC electrolyte while cells containing 2% VC+2% TAP in SL:EMC electrolyte had much better capacity retention than cells containing 2% PES in EC:EMC electrolyte. Figures 4c and 4f show that cells containing fluorinated electrolytes had similar cycling performance to cells containing 2% PES in EC:EMC electrolyte.

Figure 5a ranks the % capacity loss at cycle 400 for  $\text{LaPO}_4$ -coated NMC442/graphite pouch cells containing the different electrolyte formulations shown in Figure 3 and Figure 4. Each data point in Figure 5a represents the average of two cells and the error bars are the standard deviation of the results. From left to the right, the capacity loss increases indicating that cells on the left in Figure 5a had better capacity retention and longer cycle lifetime. Figure 5a shows that TAP containing electrolytes including 2% VC + 2% TAP in SL:EMC, 2% PES + 2% TAP in EC:EMC and 0.5% FEC + 5% TAP + 0.2%



**Figure 4.** a, b, c) Discharge capacity and d, e, f)  $\Delta V$ , all plotted vs cycle number for  $\text{LaPO}_4$ -coated NMC442/graphite pouch cells containing different electrolytes as indicated. The cycling was between 2.8 and 4.5 V at C/2.3 (80 mA) and 55°C using a continuous cycling protocol.

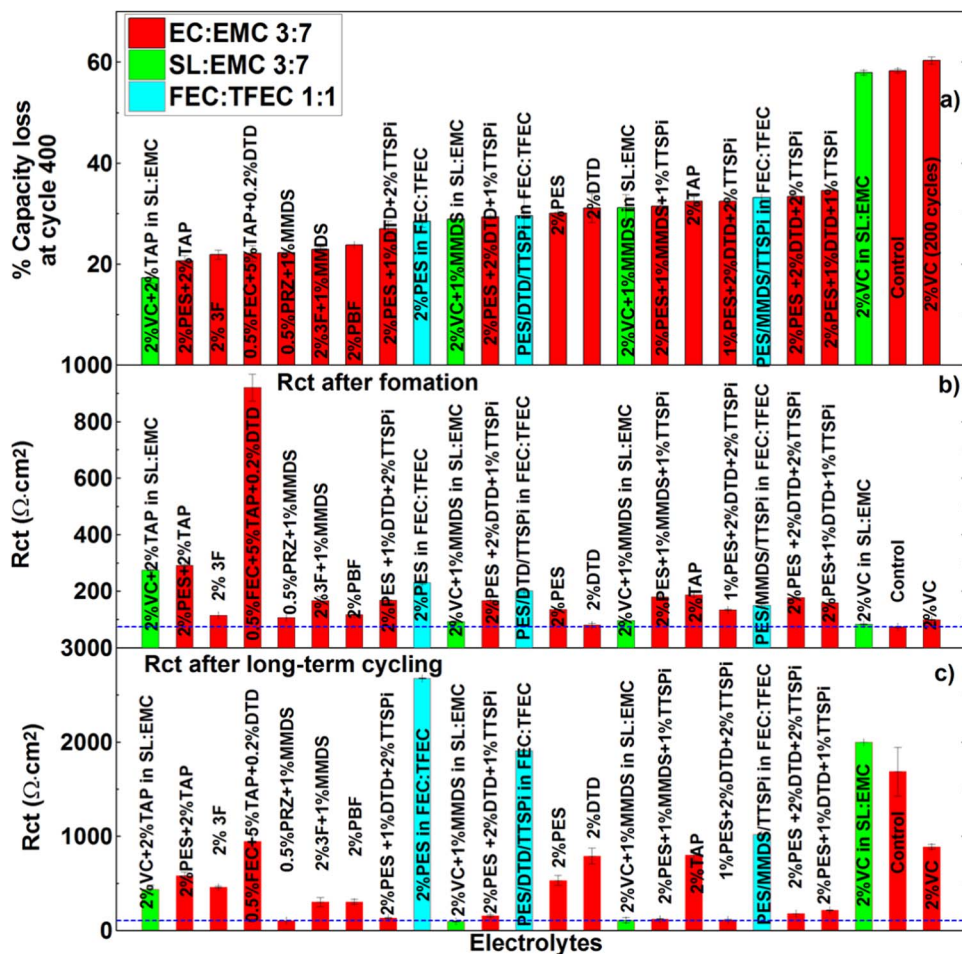
DTD in EC:EMC had the least capacity loss during 400 cycles. Figure 5a shows that cells containing PBF-based electrolyte additives or their combinations also had excellent capacity retention. Figure 5a shows that cells containing control, 2% VC in SL:EMC and 2% VC in EC:EMC had much worse capacity retention than cells containing 2% PES in EC:EMC electrolyte. Figure 5a shows that cells with the FEC-TFEC-based electrolytes had similar cycling performance to cells containing only 2% PES in EC:EMC electrolyte. These results are very different from the cycling data at 40°C reported by Xia et al.<sup>53</sup> which showed that NMC442/graphite cells with **uncoated** NMC442 electrodes and FEC-TFEC-based electrolytes had much better cycling performance than any non-fluorinated electrolytes with additives. This shows that the optimization of coatings on positive electrode materials and electrolytes must go hand in hand, a fact that may not be appreciated in the academic community.

Figure 6 shows the cycle number when cells reached 80% of their initial capacity for two large experiments. Figure 6a shows the results from Reference 53 where **uncoated** NMC442/graphite cells (240 mAh cells) were tested at 40°C and Figure 6b shows data from this paper where  $\text{LaPO}_4$ -coated NMC442/graphite cells (180 mAh cells) were tested at 55°C. Figure 6a shows that **uncoated** cells containing the FEC-TFEC-based electrolyte had the best capacity retention among the types of electrolyte tested at 40°C. Cells containing SL:EMC electrolyte with 2% VC + 2% TAP had slightly better capacity retention than the EC:EMC-based electrolyte. Figure 6b shows that TAP-containing cells had the best capacity retention while cells containing fluorinated electrolytes had similar capacity retention to cells containing the EC:EMC electrolyte with PES 211 additive blends. It is unfortunate that experiments on the uncoated NMC cells were not available at 55°C and that experiments on the coated NMC cells were not available at 40°C. Nevertheless, Figure 6 demonstrates the importance of coupling electrolyte optimization with positive electrode optimization.

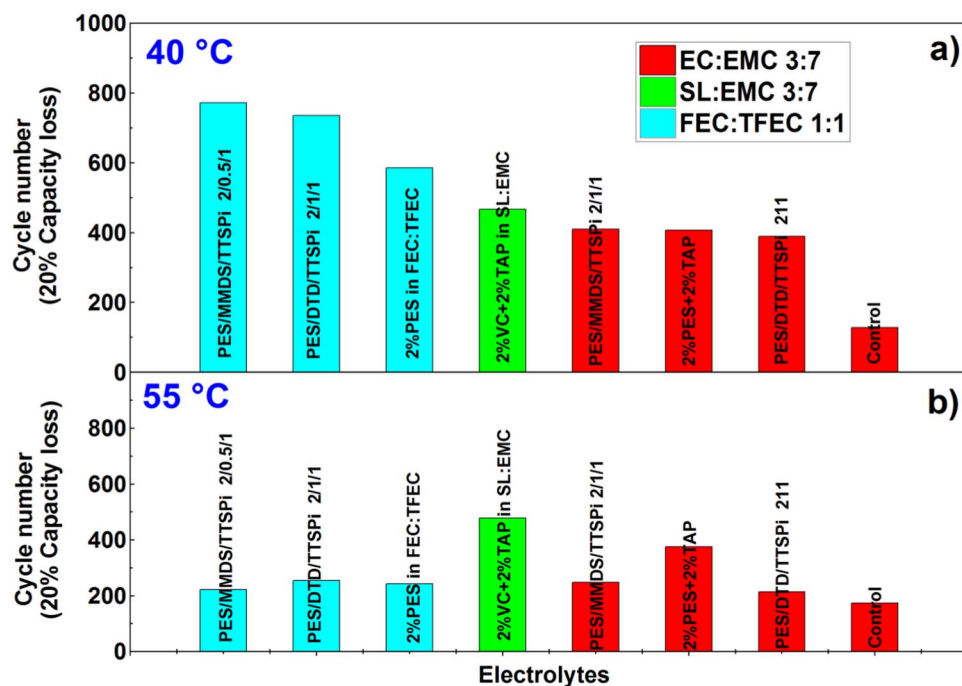
Figures 5b and 5c show the charge transfer impedance,  $R_{ct}$ , (see Figure 2d) for the same cells described in Figure 5a measured before

and after long-term cycling, respectively. Figures 5b and 5c show that the impedance in cells containing control, 2% VC in SL:EMC and 2% VC in EC:EMC electrolytes increased significantly during long-term cycling. Figures 5b and 5c show that TAP-containing cells had higher impedance after formation and that the impedance of these cells increased during long-term cycling except for cells with 0.5% FEC + 5% TAP + 0.2% DTD, which was very stable due to the large amount of TAP (high impedance) used in these cells. Figures 5b and 5c show that cells containing PBF-based electrolyte additives in EC:EMC had lower impedance both before and after cycling than cells containing 2% PES in EC:EMC electrolyte. The combination of PBF-based electrolyte additives with MMDS can further reduce the impedance during cycling. Figures 5b and 5c show that cells containing PES/DTT(MMDS)/TTSPi additive combinations had similar impedance before cycling and that the impedance of these cells were stable during the long-term cycling. Figures 5b and 5c show the impedance of cells containing FEC:TFEC electrolyte increased significantly during long-term cycling.

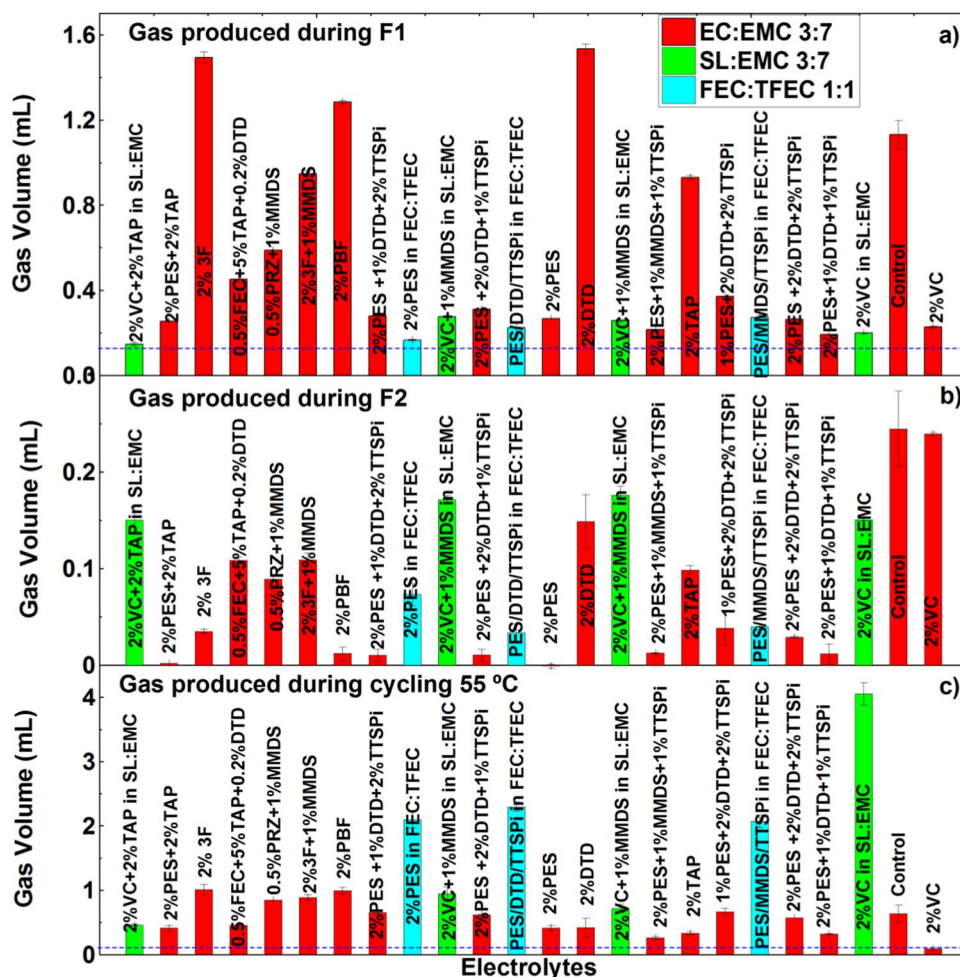
Figures 7a, 7b and 7c show the gas evolution during formation step 1 (charge to 3.5 V), during formation step 2 (charge to 4.5 V) and during long-term cycling experiments, respectively. Gas during formation step 1 is generally caused by the passivation of the graphite electrode while gas during formation step 2 is generally caused by electrolyte oxidation and the possible formation of an SEI on the NMC electrode.<sup>48</sup> Minimizing gas evolution in pouch cells during both formation and cycling is important because gas generation will lead to a loss of stack pressure in the pouch and will cause swelling and Li plating, leading to cell failure. Figure 7a shows that cells containing control, 2% DTD, 2% TAP and PBF-based electrolyte additives in EC:EMC electrolyte produced a large amount of gas during formation step 1 due to the relatively poor SEI formed on the graphite electrode. Figure 7a shows that cells containing SEI forming additives such as VC and PES produced less gas during formation step 1. Figure 7b shows the gas production during formation step 2 (charge to 4.5 V) during the first cycle. Figure 7b shows that control cells



**Figure 5.** a) The capacity loss at cycle 400 for  $\text{LaPO}_4$ -coated NMC442/graphite pouch cells containing different electrolytes with different additive sets as indicated. The capacity loss has been arranged from “lowest” in the left to “highest” in the right. The charge transfer resistance  $R_{ct}$ , measured after formation b) and after long-term cycling at  $55^\circ\text{C}$  c) The charge transfer resistance was measured at  $10^\circ\text{C}$  and 3.8 V.



**Figure 6.** The cycle number when cells lost 20% of their initial capacity. a) Data for uncoated NMC442/graphite cells tested at  $40^\circ\text{C}$  (data from Ref. 52) to 4.5 V, and b) Data for  $\text{LaPO}_4$ -coated NMC442/graphite cells from this paper tested to 4.5 V at  $55^\circ\text{C}$ .



**Figure 7.** Summary of gas evolution in  $\text{LaPO}_4$ -coated NMC442/graphite cells measured during a) formation step 1 (charge to 3.5 V at 40°C), b) formation step 2 (charge to 4.5 V at 40°C) and c) during long-term cycling at 55°C.

and the VC-containing cells produced about 0.2 mL of gas during formation step 2 while the PES-containing cells produced almost no gas. Figure 7c shows that control cells produced about 0.7 mL gas during long-term cycling while cells containing 2% VC in EC:EMC electrolyte produced about 0.1 mL gas during long-term cycling. As shown in Figure 3, cells containing 2% VC in EC:EMC electrolyte could only cycle 50 times and, therefore, the failure of cells containing 2% VC was not due to gas evolution but due to the impedance growth (Figure 5c). Figure 7c shows that TAP-containing cells (2% VC + 2% TAP in SL:EMC, 2% PES + 2% TAP in EC:EMC and 0.5% FEC + 5% TAP + 0.2% DTD in EC:EMC) produced about 0.4 mL gas while cells containing PBF-based electrolyte produced ~1 mL gas during long-term cycling. Figure 7c shows that cells containing PES/DTD(MMDS)/TTSPi produced about 0.5 mL gas during long-term cycling. Figure 7c shows that cells containing FEC:TfEC electrolyte produced more than 2 mL of gas during long-term cycling.

Figure 8 shows “radar” or “spider” plots which summarize some of the best electrolyte formulations, for the  $\text{LaPO}_4$ -coated NMC442/graphite cells operated to 4.5 V at 55°C, in this study as indicated in the legend. The four axes represent % capacity loss at cycle 400,  $\Delta V$  at cycle 400, gas evolution during long-term cycling and  $R_{ct}$  after long-term cycling ( $\Omega \cdot \text{cm}^2$ ). The axes have been scaled so that 100% corresponds to the value of the additive that has the largest (the worst) value of each parameter. Therefore, the best additive blend would have values closest to the center of the plot. Figure 8 shows that there is no clear “winner” in all aspects amongst these electrolyte formulations tested. For example, cells containing 2% VC + 2% TAP in SL:EMC electrolyte had the least capacity loss and lowest gas gener-

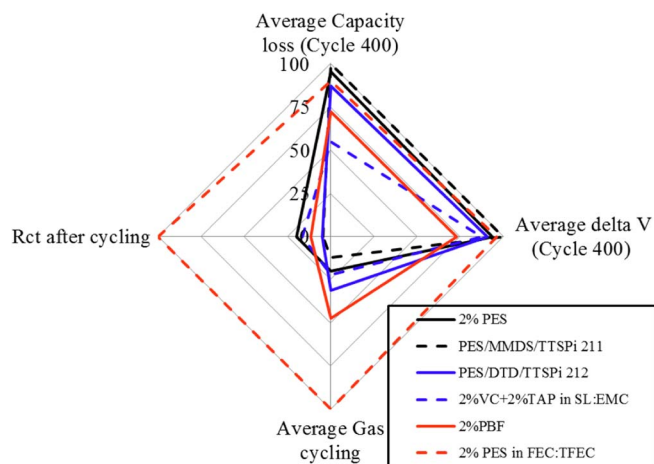
ation, however, the impedance after cycling was high, compared with PES 211 or 212 in EC:EMC electrolyte. Cells containing 2% PBF had excellent capacity retention and low impedance after cycling, however, the gas produced during cycling is large. Tradeoffs between gas, impedance and lifetime would have to be made.

## Conclusions

This work is a follow-up study to Refs. 35–47. This work compares the long-term cycling performance of these electrolyte additives and/or solvents at 4.5 V and 55°C. This work confirmed the advantage of using TAP-containing electrolyte at high temperature and high voltages. The positive effect of TAP is believed to be due to the thick SEI films formed by polymerization reactions during formation. The SEI formed in the presence of TAP is relatively stable at elevated temperature and at high voltages [41].

This work also confirmed the benefit of using PBF-containing electrolyte additives at high voltages which have been shown to be effective at impedance control in Refs. 38–40. One problem is that cells with PBF-containing electrolytes display gas evolution during formation and cycling (see Figure 7).

This work showed that the cells with PES-containing ternary electrolyte additive blends in EC:EMC electrolyte had similar cycling performance to 4.5 V to cells containing 2% PES alone in EC:EMC electrolyte. Since these PES-containing ternary electrolyte additive blends could suppress the gas evolution and  $R_{ct}$  growth at high temperature and high voltage (see Figures 5 and 7), the failure mechanism of cells with PES-containing ternary electrolyte additive blends was



**Figure 8.** Radar plots summarizing some of the results for **LaPO<sub>4</sub>-coated** NMC442/graphite cells with electrolyte formulations studied in this paper. The axes show % capacity loss at cycle 400,  $\Delta V$  at cycle 400, gas evolution during long-term cycling and  $R_{ct}$  after long-term cycling ( $\Omega \cdot \text{cm}^2$ ). The axes have been scaled so that 100% is the value of the additive that has the largest (the worst) value of each parameter. Therefore, the best additive blend would have values closest to the center of the plot.

most likely caused by other forms of impedance growth such as contact impedance ( $\Delta V$  increases in Figure 3).

Refs. 47 and 53 show that **uncoated** NMC442/graphite cells with FEC-TFEC-based electrolytes had much better cycling performance than the non-fluorinated electrolytes at 4.5 V and 40°C. However, the work here shows that **LaPO<sub>4</sub>-coated** NMC442/graphite cells with FEC:TFEC-based electrolyte had similar cycling performance to cells containing only 2% PES in EC:EMC electrolyte at 55°C. The main problems with FEC-TFEC-based electrolytes at high temperatures were gas evolution and impedance growth (Figures 5 and 7), leading to capacity loss. The gassing problem of FEC:TFEC-based electrolytes may be caused by the large amounts of FEC used which was apparently not stable at high temperature and/or high potentials<sup>6,54</sup>.

Successfully operating Li-ion cells to high voltages at high temperatures may require the use a combination of new solvent blends, additives as well as proper coating materials. This may take years of searching and optimization. We encourage others to be involved in this search.

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